

CHAPTER 19

Chemical Coordination and Integration

Hypothalamus, Pituitary Gland, Pineal Gland

- Hormones stored and released from neurohypophysis are: *(2020-Covid)*
 - Oxytocin and Vasopressin
 - Follicle stimulating hormone and Luteinizing hormone
 - Prolactin and Vasopressin
 - Thyroid stimulating hormone and oxytocin
- Hypersecretion of growth hormone in adults does not cause further increase in height, because *(2017-Delhi)*
 - Growth hormone becomes inactive in adults
 - Epiphyseal plates close after adolescence
 - Bones lose their sensitivity to growth hormone in adults
 - Muscle fibres do not grow in size after birth
- GnRH, a hypothalamic hormone, needed in reproduction, acts on *(2017-Delhi)*
 - Anterior pituitary gland and stimulates secretion of LH and oxytocin
 - Anterior pituitary gland and stimulates secretion of LH and FSH
 - Posterior pituitary gland and stimulates secretion of oxytocin and FSH
 - Posterior pituitary gland and stimulates secretion of LH and relaxin
- The posterior pituitary gland is not a 'true' endocrine gland because: *(2016 - II)*
 - It is under the regulation of hypothalamus
 - It secretes enzymes
 - It is provided with a duct
 - It only stores and releases hormones
- Which one of the following hormones though synthesised elsewhere is stored and released by the master gland? *(2015 Re)*
 - Luteinising hormone
 - Prolactin
 - Melanocyte stimulating hormone
 - Antidiuretic hormone

Thyroid, Parathyroid, Thymus

- Which of the following are NOT under the control of thyroid hormone? *(2023)*
 - Maintenance of water and electrolyte balance
 - Regulation of basal metabolic rate
 - Normal rhythm of sleep-wake cycle
 - Development of immune system
 - Support the process of R.B.Cs formation

Choose the correct answer from the options given below:

 - D and E only
 - A and D only
 - B and C only
 - C and D only
- Which of the following statements are **correct** with respect to the hormone and its function? *(2023-Manipur)*
 - Thyrocalcitonin (TCT) regulates the blood calcium level.
 - In males, FSH and androgens regulate spermatogenesis.
 - Hyperthyroidism can lead to goitre.
 - Glucocorticoids are secreted in Adrenal Medulla.
 - Parathyroid hormone is regulated by circulating levels of sodium ions.

Choose the **most appropriate** answer from the options given below:

 - (C) and (E) only
 - (A) and (B) only
 - (B) and (C) only
 - (A) and (D) only
- Given below are two statements: *(2023-Manipur)*

Statement-I: Parathyroid hormone acts on bones and stimulates the process of bone resorption.

Statement-II: Parathyroid hormone along with Thyrocalcitonin plays a significant role in carbohydrate metabolism.

In the light of the above statements, choose the **correct** answer from the options given below:

 - Statement-I** is correct but **Statement-II** is incorrect
 - Statement-I** is incorrect but **Statement-II** is correct
 - Both **Statement-I** and **Statement-II** are correct.
 - Both **Statement-I** and **Statement-II** are incorrect.

9. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R). (2022 Re)

Assertion (A): During pregnancy the level of thyroxine is increased in the maternal blood.

Reason (R): Pregnancy is characterised by metabolic changes in the mother.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) is not correct but (R) is correct
 - Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (A) is correct but (R) is not correct
10. Which of the following are **not** the effects of Parathyroid hormone? (2022)

- Stimulates the process of bone resorption
- Decrease Ca^{2+} level in blood
- Reabsorption of Ca^{2+} by renal tubules
- Decrease the absorption of Ca^{2+} from digested food
- Increases metabolism of carbohydrates

Choose the most appropriate answer from the options given below:

- B and C only
 - A and C only
 - B, D and E only
 - A and E only
11. Which of the following hormones can play a significant role in osteoporosis? (2018)
- Aldosterone and Prolactin
 - Progesterone and Aldosterone
 - Estrogen and Parathyroid hormone
 - Parathyroid hormone and Prolactin

12. Thymosin is responsible for: (2017-Delhi)

- Decreased production of T-lymphocytes
- Inhibiting the production of antibodies
- Decreasing the blood calcium level in old individuals
- Increased production of T-lymphocytes

13. Identify the hormone with its correct matching of source and function: (2014)

- Atrial natriuretic factor: Ventricular wall, increases the blood pressure
- Oxytocin: Posterior pituitary, growth and maintenance of mammary glands
- Melatonin: Pineal gland, regulates the normal rhythm of sleep wake cycle
- Progesterone: Corpus luteum, stimulation of growth and activities of female secondary sex organs

14. A pregnant female delivers a baby who suffers from stunted growth, mental retardation, low intelligence quotient and abnormal skin. This is the result of: (2013)

- Over-secretion of pars distalis
- Deficiency of iodine in diet
- Low secretion of growth hormone
- Cancer of the thyroid gland

Adrenal gland, Pancreas

15. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Epinephrine	(I)	Hyperglycemia
(B)	Thyroxine	(II)	Smooth muscle contraction
(C)	Oxytocin	(III)	Basal metabolic rate
(D)	Glucagon	(IV)	Emergency hormone

Choose the **correct** answer from the options given below:

- A-(II), B-(I), C-(IV), D-(III)
- A-(III), B-(II), C-(I), D-(IV)
- A-(IV), B-(III), C-(II), D-(I)
- A-(I), B-(IV), C-(III), D-(II)

16. Match List-I with List-II: (2024)

List-I		List-II	
(A)	Exophthalmic goitre	(I)	Excess secretion of cortisol, moon face and hyperglycemia
(B)	Acromegaly	(II)	Hypo-secretion of thyroid hormone and stunted growth
(C)	Cushing's syndrome	(III)	Hyper secretion of thyroid hormone and protruding eye balls.
(D)	Cretinism	(IV)	Excessive secretion of growth hormone

Choose the correct answer from the options given below:

- A-(III), B-(IV), C-(II), D-(I)
- A-(III), B-(IV), C-(I), D-(II)
- A-(I), B-(III), C-(II), D-(IV)
- A-(IV), B-(II), C-(I), D-(III)

17. Presence of which of the following conditions in urine are indicative of Diabetes Mellitus? (2020)

- Uremia and Renal Calculi
- Ketonuria and Glycosuria
- Renal calculi and Hyperglycaemia
- Uremia and Ketonuria

18. Match the following columns and select the correct option. (2020)

	Column-I		Column-II
1.	Pituitary gland	(i)	Grave's disease
2.	Thyroid gland	(ii)	Diabetes mellitus
3.	Adrenal gland	(iii)	Diabetes insipidus
4.	Pancreas	(iv)	Addison's disease

(1) (2) (3) (4) (1) (2) (3) (4)

- (iii) (ii) (i) (iv) b. (iii) (i) (iv) (ii)
- (ii) (i) (iv) (iii) d. (iv) (iii) (i) (ii)

19. Which of the following would help in prevention of diuresis? (2020)

- Reabsorption of Na^+ and water from renal tubules due to aldosterone
- Atrial natriuretic factor causes vasoconstriction
- Decrease in secretion of renin by JG cells
- More water reabsorption due to undersecretion of ADH

20. Select the correct statement (2020)

- Glucagon is associated with hypoglycemia.
- Insulin acts on pancreatic cells and adipocytes.
- Insulin is associated with hyperglycemia.
- Glucocorticoids stimulate gluconeogenesis.

21. Match the following hormones with the respective disease (2019)

List-I		List-II	
A.	Insulin	i.	Addison's disease
B.	Thyroxin	ii.	Diabetes insipidus
C.	Corticoids	iii.	Acromegaly
D.	Growth Hormone	iv.	Goitre
		v.	Diabetes mellitus

Select the correct option.

- (A) (B) (C) (D) (A) (B) (C) (D)
- (v) (i) (ii) (iii) b. (ii) (iv) (iii) (i)
 - (v) (iv) (i) (iii) d. (ii) (iv) (i) (iii)

22. Graves' disease is caused due to: (2016 - II)

- Hyposecretion of adrenal gland
- Hypersecretion of adrenal gland
- Hyposecretion of thyroid gland
- Hypersecretion of thyroid gland

23. A chemical signal that has both endocrine and neural roles is: (2015)

- Epinephrine
- Cortisol
- Melatonin
- Calcitonin

24. Which one of the following hormones is not involved in sugar metabolism? (2015 Re)

- Aldosterone
- Insulin
- Glucagon
- Cortisone

25. Fight-or-flight reactions cause activation of: (2014)

- The pancreas leading to a reduction in the blood sugar levels
- The parathyroid glands, leading to increased metabolic rate
- The kidney, leading to suppression of renin angiotensin-aldosterone pathway
- The adrenal medulla, leading to increased secretion of epinephrine and norepinephrine

Testis, Ovary, Hormones of Heart, Kidney and GI Tract

26. Identify the wrong statements: (2024 Re)

- Erythropoietin is produced by juxtaglomerular cells of the kidney
- Leydig cells produce Androgens
- Atrial Natriuretic factor, a peptide hormone is secreted by the seminiferous tubules of the testes

D. Cholecystokinin is produced by gastro-intestinal tract

E. Gastrin acts on intestinal wall and helps in the production of pepsinogen

Choose the **most appropriate** answer from the options given below:

- D and E only
- A and B only
- C and E only
- A and C only

27. Match List-I with List-II. (2023)

List-I		List-II	
(A)	CCK	(I)	Kidney
(B)	GIP	(II)	Heart
(C)	ANF	(III)	Gastric gland
(D)	ADH	(IV)	Pancreas

Choose the correct answer from the options given below:

- A-IV, B-II, C-III, D-I
- A-IV, B-III, C-II, D-I
- A-III, B-II, C-IV, D-I
- A-II, B-IV, C-I, D-III

28. Erythropoietin hormone which stimulates R.B.C. formation is produced by: (2021)

- The cells of rostral adenohypophysis
- The cells of bone marrow
- Juxtaglomerular cells of the kidney
- Alpha cells of pancreas.

29. Match the following columns and select the correct option: (2020-Covid)

Column-I		Column-II	
1.	Ovary	(i)	Human chorionic Gonadotropin
2.	Placenta	(ii)	Estrogen & Progesterone
3.	Corpus luteum	(iii)	Androgens
4.	Leydig cells	(iv)	Progesterone only

Select the correct option from following:

- (1) (2) (3) (4)
- (i) (ii) (iii) (iv)
 - (i) (iii) (ii) (iv)
 - (ii) (i) (iv) (iii)
 - (iv) (iii) (ii) (i)

30. Which of the following pairs of hormones are not antagonistic (having opposite effects) to each other? (2016 - I)

a.	Parathormone	1.	Calcitonin
b.	Insulin	2.	Glucagon
c.	Aldosterone	3.	Atrial Natriuretic Factor
d.	Relaxin	4.	Inhibin

31. Which hormones do stimulate the production of pancreatic juice and bicarbonate? (2016 - II)

- Cholecystokinin and secretin
- Insulin and glucagon
- Angiotensin and epinephrine
- Gastrin and insulin

32. Which of the following statement is correct in relation to the endocrine system? (2013)
- Releasing and inhibitory hormones are produced by the pituitary gland
 - Adenohypophysis is under direct neural regulation of the hypothalamus
 - Organs in the body like gastrointestinal tract, heart, kidney and liver do not produce any hormones
 - Non - nutrient chemicals produced by the body in trace amount that act as intercellular messenger are known as hormones
33. Which one of the following is not the function of placenta? (2013)
- Secretes oxytocin during parturition
 - Facilitates supply of oxygen and nutrients to embryo
 - Secretes estrogen
 - Facilitates removal of carbon dioxide and waste material from embryo

Mechanism of Hormone Action

34. Which of the following is not a steroid hormone? (2024)
- Progesterone
 - Glucagon
 - Cortisol
 - Testosterone
35. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**. (2022 Re)
- Assertion (A):** FSH which interacts with membrane bound receptors does not enter the target cell.
- Reason (R):** Binding of FSH to its receptors generates second messenger (cyclic AMP) for its biochemical and physiological responses.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (A) is not correct but (R) is correct
 - Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (A) is correct but (R) is not correct

36. Match the following columns and select the correct option: (2020-Covid)

	Column-I		Column-II
1.	Pituitary hormone	(i)	Steroid
2.	Epinephrine	(ii)	Neuropeptides
3.	Endorphins	(iii)	Peptides, proteins
4.	Cortisol	(iv)	Biogenic amines

- (1) (2) (3) (4)
- (iii) (iv) (ii) (i)
 - (iv) (iii) (i) (ii)
 - (iii) (iv) (i) (ii)
 - (iv) (i) (ii) (iii)
37. How does steroid hormone influence the cellular activities? (2019)
- Changing the permeability of the cell membrane
 - Binding to DNA and forming a gene-hormone complex
 - Activating cyclic AMP located on the cell membrane
 - Using aquaporin channels as second messenger
38. Which of the following is an amino acid derived hormone? (2018)
- Epinephrine
 - Ecdysone
 - Estradiol
 - Estriol
39. Name a peptide hormone which acts mainly on hepatocytes, adipocytes and enhances cellular glucose uptake and utilisation. (2016 - II)
- Secretin
 - Gastrin
 - Insulin
 - Glucagon
40. The amino acid Tryptophan is the precursor for the synthesis of: (2016 - I)
- Melatonin and Serotonin
 - Thyroxine and Triiodothyronine
 - Estrogen and Progesterone
 - Cortisol and Cortisone

Answer Key

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (b) | 4. (d) | 5. (d) | 6. (d) | 7. (b) | 8. (a) | 9. (b) | 10. (c) |
| 11. (c) | 12. (d) | 13. (c) | 14. (b) | 15. (c) | 16. (b) | 17. (b) | 18. (b) | 19. (a) | 20. (d) |
| 21. (c) | 22. (d) | 23. (a) | 24. (a) | 25. (d) | 26. (c) | 27. (b) | 28. (c) | 29. (c) | 30. (d) |
| 31. (a) | 32. (d) | 33. (a) | 34. (b) | 35. (c) | 36. (a) | 37. (b) | 38. (a) | 39. (c) | 40. (a) |